

In Hilb, Frobenius implies the H^* axiom

The Hilbert-space branch of arXiv:1011.6123 is answered by arXiv:2003.04149

Literature-status identification, lane 12 · 13 August 2026

Status. `literature_already_answered` for the central category **Hilb**. Poinsoot proves the required semisimplicity, and in fact proves a stronger equivalence without the source’s specialness axiom. The source’s separate question for arbitrary monoidal dagger categories is not claimed here.

The source question

Abramsky and Heunen consider a bounded comultiplication $\delta : H \rightarrow H \otimes H$ satisfying associativity (A), commutativity (C), specialness (M), and the Frobenius law (F). They ask whether these axioms force their H^* axiom (H), first in **Hilb** and then in general monoidal dagger categories [1]. Writing

$$\mu = \delta^\dagger : H \otimes H \longrightarrow H,$$

axiom (M) is $\delta^\dagger \delta = I_H$, equivalently $\mu \mu^\dagger = I_H$. Their structure theorem shows that, in **Hilb**, (H) is equivalent to semisimplicity; they consequently reformulate the question as whether a nonzero radical Frobenius algebra exists.

Decisive later theorem

Poinsoot calls (H, μ) a commutative Hilbertian Frobenius semigroup when μ is a bounded commutative associative multiplication and $\mu^\dagger$ is a morphism of H -bimodules. This is exactly the operator form of (A), (C), and (F). Theorem 33 of [2] proves, for every such semigroup, the equivalence

$$\text{semisimple} \iff \mu \mu^\dagger \text{ is injective.}$$

Under (M) the latter operator is the identity. Hence the algebra is semisimple, so the equivalence established in arXiv:1011.6123 gives (H). Equivalently, Corollary 32 of [2] says that a radical commutative Hilbertian Frobenius semigroup must have $\mu = 0$; this cannot satisfy $\mu \mu^\dagger = I_H$ unless $H = 0$. Thus no nontrivial radical example of the kind sought by the source exists.

Poinsoot explicitly records immediately after Theorem 33 that the special case answers the main **Hilb** question of Abramsky–Heunen. Proposition 39 goes further: a commutative Hilbertian algebra is a Hilbertian H^* -algebra if and only if it is a Hilbertian Frobenius semigroup, without assuming specialness.

Scope

The argument is operator-theoretic and is stated for Hilbert spaces. It settles both equivalent **Hilb** formulations in the source—(F) $\Rightarrow$ (H) under (A),(C),(M), and nonexistence of a nontrivial special radical Frobenius algebra. It does not by itself establish the corresponding implication in an arbitrary monoidal dagger category. A bounded exact-phrase and axiom-based search found no source justifying such a general promotion.

References

- [1] S. Abramsky and C. Heunen, *H^* -algebras and nonunital Frobenius algebras: first steps in infinite-dimensional categorical quantum mechanics*, Mathematical Structures in Computer Science 22 (2012), 1–24; [arXiv:1011.6123](#).
- [2] L. Poinsoot, *Hilbertian Frobenius algebras*, Communications in Algebra 50 (2022), 85–112; [arXiv:2003.04149](#).